

Front End for Recording on PC from Audio Tapes and Old Disks



RAJ K. GORKHALI

Like to try using your personal computer (PC) for CD-quality digital audio recording? Or for making your own CD copies of treasured old analogue recordings? Most modern PCs and their sound cards are capable of doing this, but it will be a lot easier if you build and use this handy recording front end. It provides stereo preamps for a magnetic pickup and microphone, combined with basic mixing and tone control facilities.

One very attractive use for such a system is for making personal CD copies of old and treasured analogue recordings from an old shellac recording disk, or 78/45/33.3rpm commercial pressing, or even old reel-to-reel magnetic tapes. As part of the copying process you can often take the opportunity to 'clean up' the old recordings, by removing irritating imperfections such as surface noise and crackle.

Circuit and working

The unit consists of four sections: a stereo magnetic pickup preamp (IC1) with appropriate gain and RIAA equalisation for conditioning the signals from a magnetic pickup; stereo mic preamp (IC2) with sufficient gain to operate with most common dynamic or electret microphones; simple 2-input stereo mixer (IC3) to allow control of signals from the mic preamp, magnetic pickup preamp or tape/line level stereo input; and an active tone-control circuit (IC4) with bass and treble controls. It is powered by a 12V DC source. The unit with the PC and its sound card makes a convenient digital audio recording system.

The magnetic pickup preamp (IC1) is of standard design. Feedback arrangements (R12-R15, C12-C14) give a very close approximation of the

correct RIAA equalisation characteristics, to ensure a clean and balanced output from the majority of magnetic cartridges (moving-magnet type).

The mic preamp stage (IC2) is quite conventional. Negative feedback is used to determine voltage gain. Switch SW2A/B allows the gain of each stage to be set to either high or low, to suite either low output or higher output microphones.

Switch SW1A/B allows selection of either output of the magnetic pickup (PU) preamp (IC1) or those at tape/line level inputs as inputs for PU/tape channel of the mixer, controlled by ganged volume pots VR1A/B. Outputs from the mic preamp (IC2) are taken directly to mic channel volume pots VR2A/B.

Mixing stage (IC3) is again very conventional, with resistors R25-R46 and R28-R30 being used to achieve low interaction virtual earth mixing of the two input channels.

Tone control stage (IC4) is also of conventional design. The mixing and tone control circuitry have the ability to provide over 6V p-p output before clipping into the line input circuitry of a typical PC sound card.

The op-amps are operated off regulated $\pm 5.5V$ supply rails. These are derived from the nominal 12V DC input (CON7) by a simple onboard power supply circuit. Regulator IC 7805 (IC6) provides +5.5V rail directly, while the negative rail is generated by a simple polarity-inverting circuit, which uses 555 timer IC5 as a self-oscillating power chopper driving C40/C41 and D2/D3 as a charge-pump rectifier driving negative regulator IC 7905 (IC7). Negative voltage across C41 is around 9.5V.

IC6 and IC7 have 1N914 (or similar) diodes in series with their common lead to increase their outputs by about 0.5V.

PARTS LIST

Semiconductors:

IC1-IC4	- LM833 dual-audio operational amplifier
IC5	- NE555 precision timer
IC6	- 7805 positive 5V voltage regulator
IC7	- 7905 negative 5V voltage regulator
D1-D3	- 1N4001 general-purpose diode
D4, D5	- 1N914 fast-switching diode
LED1	- 3mm LED

Resistors (all 1/4-watt, $\pm 1\%$ carbon):

R1, R5, R30, R27	- 100-kilo-ohm
R2, R6	- 180-ohm
R3, R7, R35-R37	- 27-kilo-ohm
R42-R44	- 68-kilo-ohm
R4, R8, R11, R19	- 150-ohm
R9, R17	- 180-kilo-ohm
R10, R18	- 82-ohm
R12, R20	- 3.9-kilo-ohm
R13, R21	- 39-ohm
R14, R22	- 47-kilo-ohm
R15, R23, R25, R26, R28, R29	- 470-kilo-ohm
R16, R24, R41, R48	- 22-kilo-ohm
R31, R32	- 4.7-kilo-ohm
R38, R39, R45, R46	- 2.7-kilo-ohm
R47	- 470-ohm
R50-R52	- 50k ganged logarithmic potentiometer
VR1A, VR1B	- 50k ganged linear potentiometer
VR2A, VR2B	- 100k ganged linear potentiometer
VR3A, VR3B	- 25k ganged linear potentiometer
VR4A, VR4B	- 25k ganged linear potentiometer

Capacitors:

C1, C2, C4-C7, C9, C16	- 47 μ F, 16V electrolytic
C3, C8, C23, C27, C35, C36	- 0.1 μ F ceramic disk
C10, C17	- 100 μ F ceramic disk
C11, C18	- 2.2 μ F 25V electrolytic
C19, C40, C41	- 220 μ F, 16V electrolytic
C13, C20	- 1nF ceramic disk
C14, C21	- 68nF ceramic disk
C15, C22, C33, C34	- 0.47 μ F ceramic disk
C24, C29, C39	- 1nF ceramic disk
C25, C26, C30, C31	- 4.7nF ceramic disk
C28, C32	- 220pF ceramic disk
C37	- 1000 μ F, 25V electrolytic
C38	- 3.3nF ceramic disk
C42, C43	- 100 μ F, 16V electrolytic

Miscellaneous:

L1, L2	- 8T of 0.25mm ECW on F29 bead
L3, L4	- 1mH decoupling choke
SW1A, SW1B, SW2A, SW2B	- DPDT switch
CON1-CON4, CON6, CON8	- 2-pin connector
CON5	- RCA socket
CON7	- 3-pin connector
	- Stereo dynamic microphone
	- 2-pin connector
	- Power jack